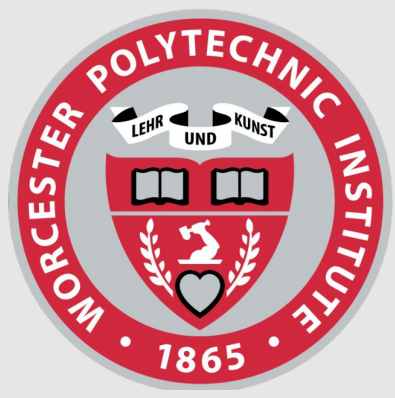


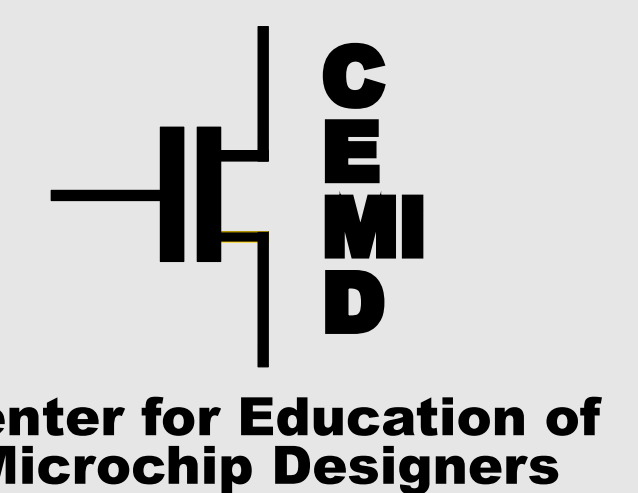
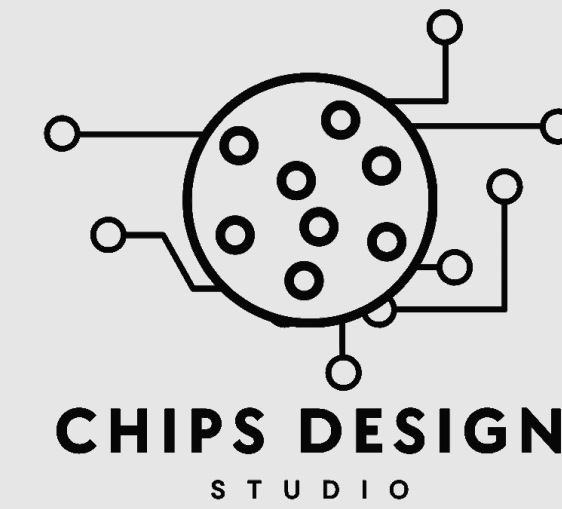


WPI

CRIMSON Power Management IC

Leo Borsari (Chip Lead), Diego Canovas, Peter Danise, Dennis Garvey, Ilia Pavliqi, Luke Ramsay, Luca Tropea, Cameron Wilding
Advisors: Dr. Suat Ay, Ibrahim Bozvel

Sponsors: UCLA Center for the Education of Microchip Designers (CEMiD), WPI-ECE CHIPS Design Studio



Abstract

We created the Conversion and Regulation Integrated Management Solution (CRIMSON), which is an intelligent power management integrated circuit (PMIC) design with **full user configurability**. This project provided WPI seniors the opportunity to work with **state-of-the-art design principles** and to gain **competitive analog IC design** experience. This design will be manufactured by TSMC in June 2026 using the TSMC 180nm CMOS 3v3/1v8, 1P6M process.

IC Design at WPI in 2026

We designed CRIMSON using two **industry-standard** IC design tools. All analog design, layout, and simulations were performed in **Cadence Virtuoso**; logic simulations and schematic generation were performed in **AMD Vivado**, then recreated in Virtuoso environment using **real TSMC180nm components**.

Our funding from the **UCLA Center for the Education of Microchip Designers (CEMiD)** affords us the ability to reserve several slots on a TSMC silicon wafer through **Muse Semiconductor**. We will submit our final design for fabrication in June 2026.

This project was designed under supervision from the **WPI CHIPS Design Studio**, managed by Dr. Suat Ay and Ibrahim Bozvel. The CHIPS Design Studio is one of several organizations on campus that serve students interested in IC design. Others include the ICAS Lab managed by Professor Ulkuhan Guler, and the IEEE Solid-State Circuit Society (SSCS) Student Chapter at WPI, the internationally recognized IEEE organization.

Purpose

IoT & Edge Devices

CRIMSON was designed for usage primarily in **IoT/edge systems**. An edge device serves as the intelligent boundary between the physical environment and the digital network, commonly acting as a sensing node in applications that require **remote monitoring** of environmental data. Edge devices are **typically limited in their computing resources**, and are often subject to harsh conditions. The combination of harsh conditions and limited computing resources makes it difficult to reliably monitor the environment while ensuring the device is operating properly. The **CRIMSON Power Monitoring and Management Architecture** gives it the **unique** ability to **selectively self-monitor** and **self-regulate** available power. This unique ability offloads the computing necessary to ensure the gadget is receiving adequate power while in the field.

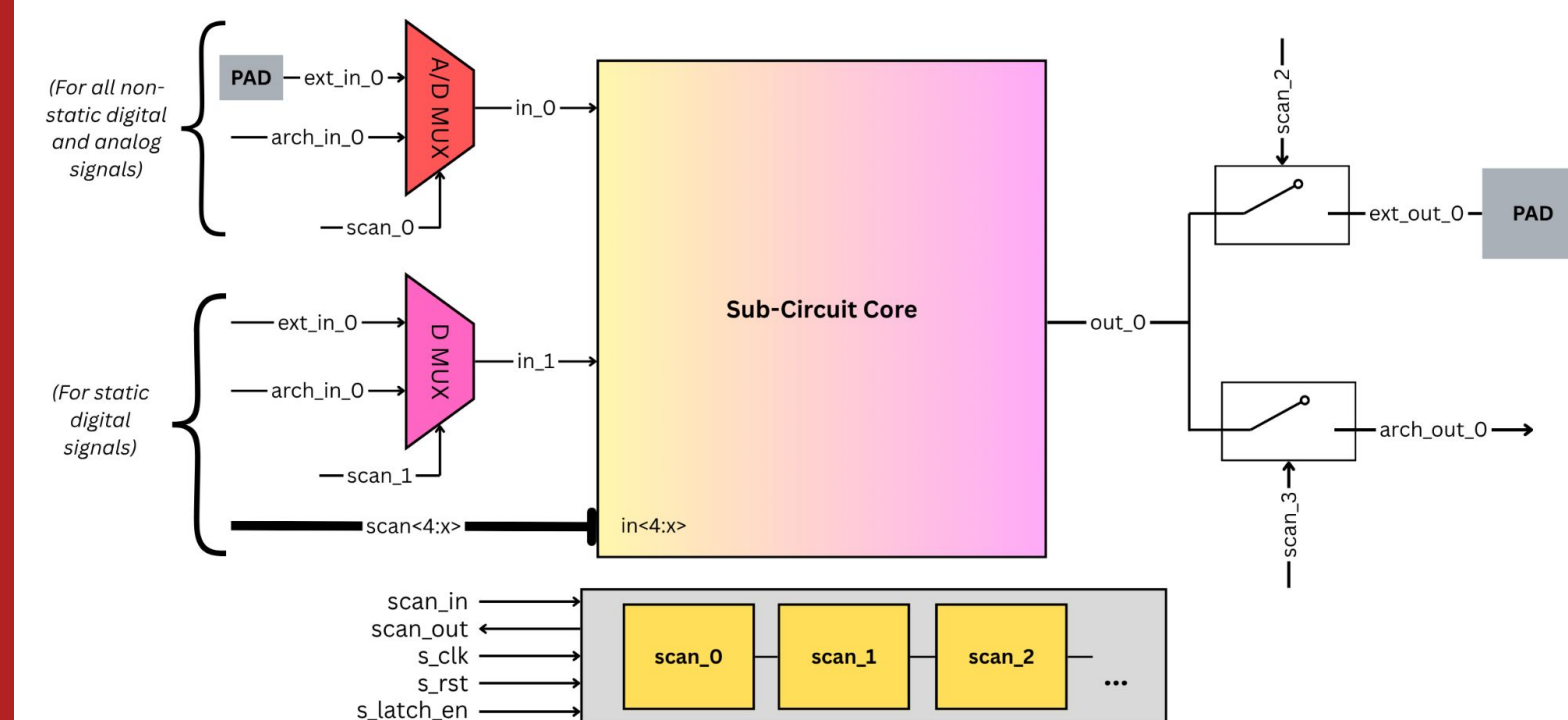
Full User Configurability

The **Configurability (CFG) Layer** is a series of **programmable interconnects** on the chip, which allows the user to selectively enable/disable all inputs and outputs of each block. Using the CFG Layer, the user **programmatically tailors** CRIMSON to their specific use-case. They can choose to design their own connections, or they can use the default use case defined in the **Wiring Diagram**, and **Power and Monitoring Architectures**. The CFG Layer also allows each block on CRIMSON to be tested and characterized by future MQP's.

Keeping WPI On Top

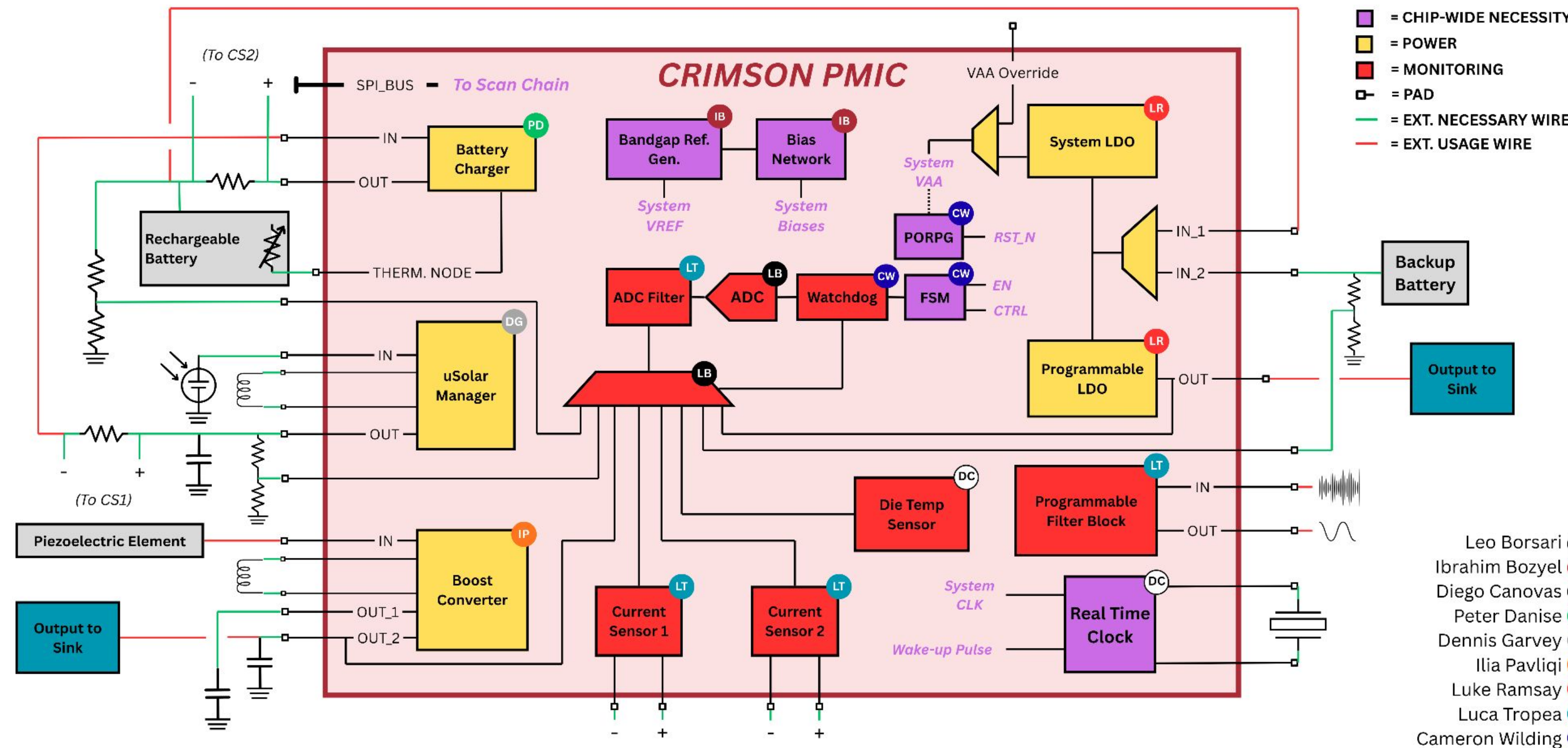
Our design uses several pre-existing components from the CHIPS Design Studio, and contributes many more to expand the Lab's portfolio. CRIMSON helps continue to lay the **groundwork for undergraduate IC design** at WPI, allowing WPI to remain at the top of the **most competitive and lucrative** engineering disciplines.

CRIMSON Configurability Layer



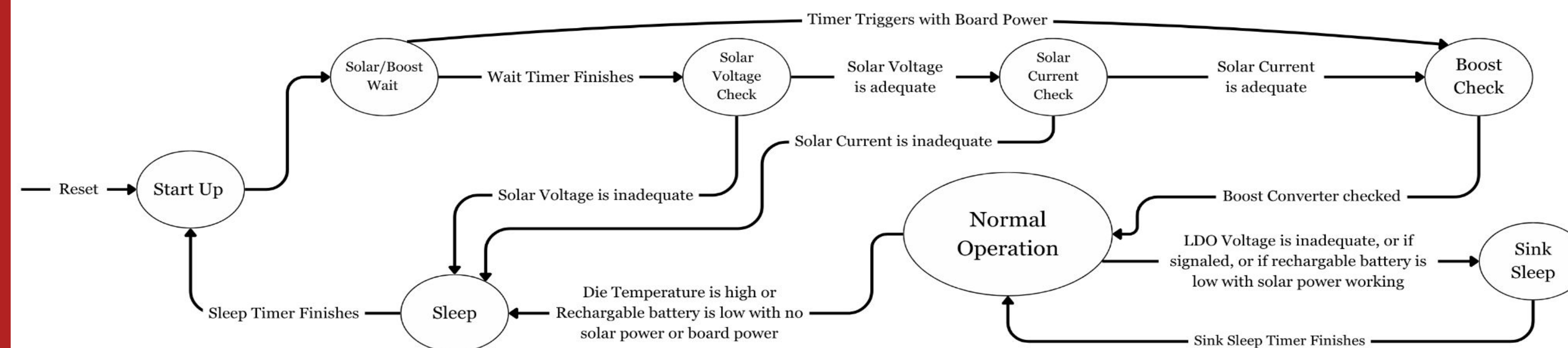
- ❖ Programmable, **reconfigurable interconnects** using Scan Chain and MUXes.
- ❖ Enables **complete isolation** of each block.
- ❖ The Configurability Layer is the pathway to **total user customization**.

CRIMSON Wiring Diagram

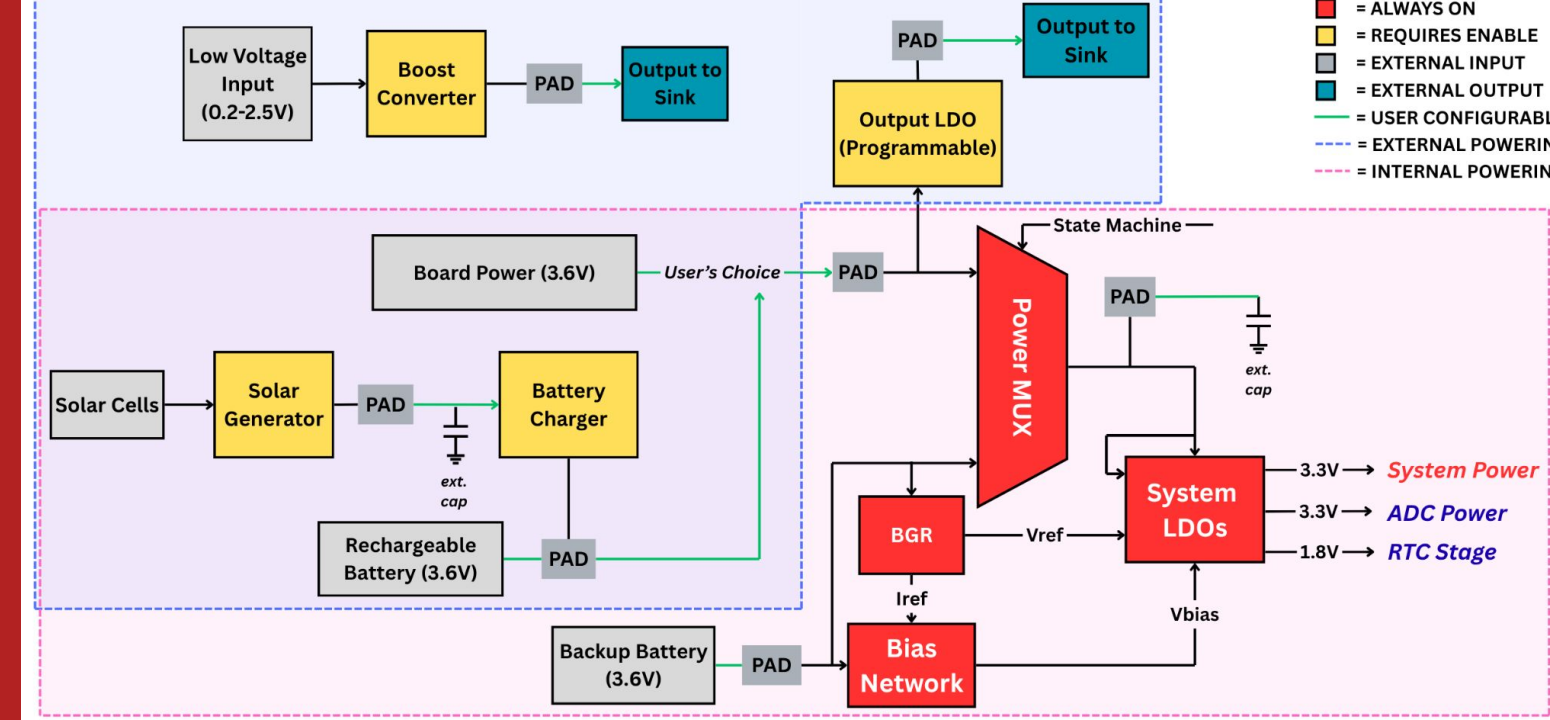


Leo Borsari (LB)
Ibrahim Bozvel (IB)
Diego Canovas (DC)
Peter Danise (PD)
Dennis Garvey (DG)
Ilia Pavliqi (IP)
Luke Ramsay (LR)
Luca Tropea (LT)
Cameron Wilding (CW)

Sequence Diagram

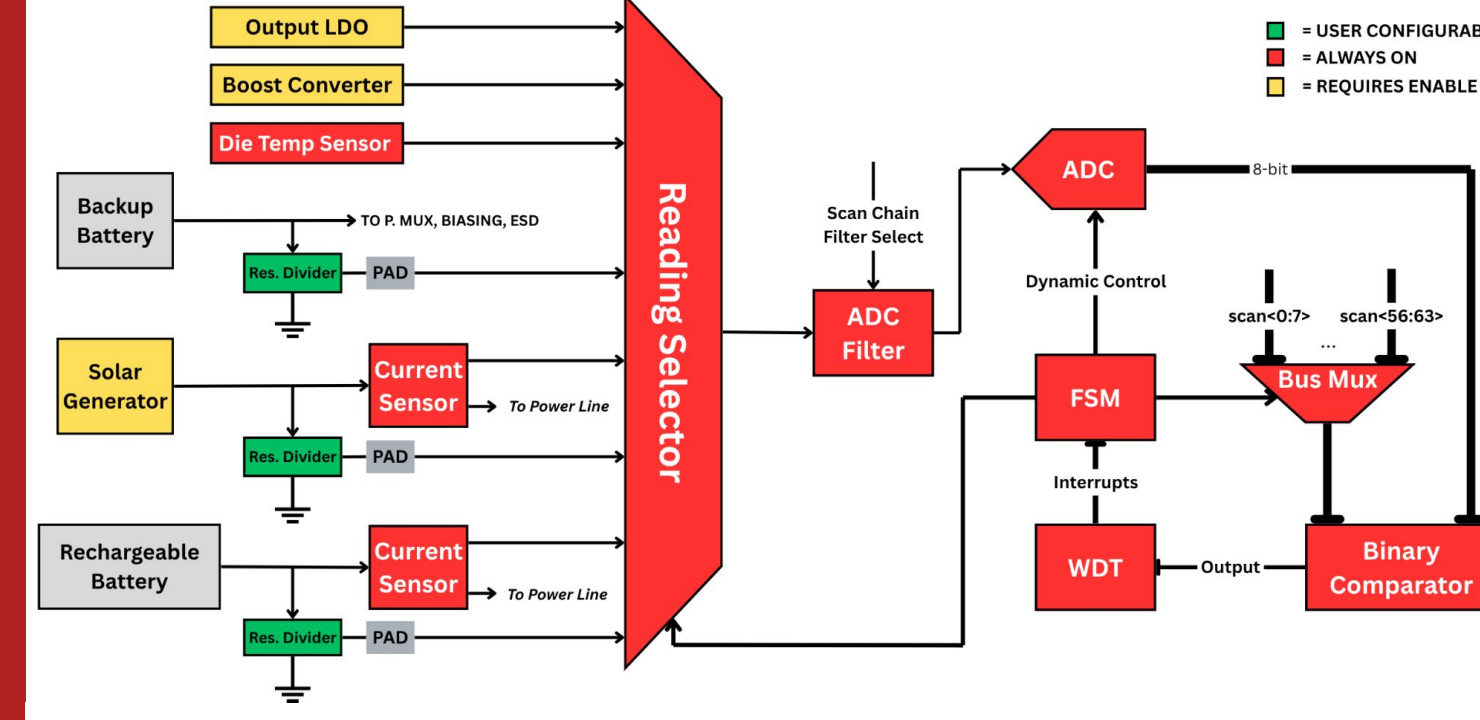


CRIMSON Power Architecture



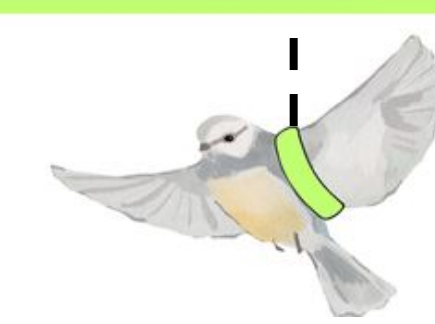
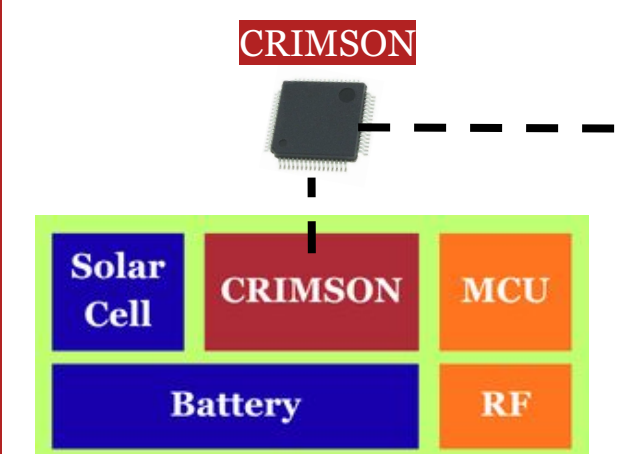
- ❖ The Power Architecture is part of the **CRIMSON Intended Usage (CIU)**.
- ❖ The architecture is designed to allow CRIMSON to **harvest energy** to provide sufficient power for itself and other off-chip circuits/systems.

CRIMSON Monitoring Architecture



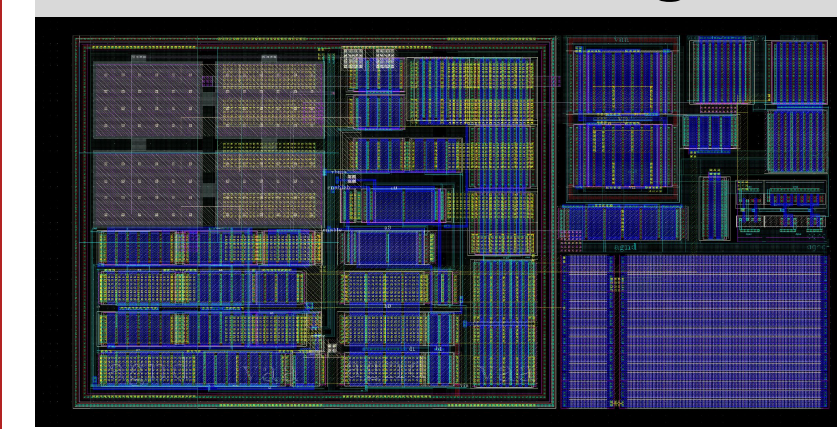
- ❖ The second part of the CIU, the Monitoring Architecture allows CRIMSON to perform **self-monitoring and self-regulation**.
- ❖ On-chip monitoring allows CRIMSON to **excel in edge-device applications** by **reducing the required compute workload** on the off-chip MCU.

TQFP-64 Package: 10mm x 10mm (Actual Size)

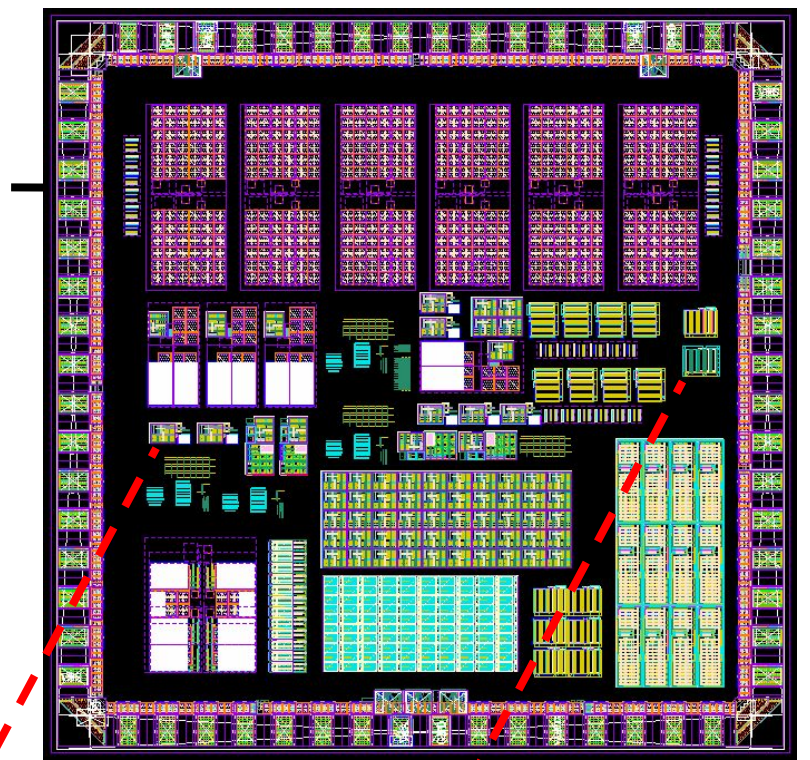


Bird Tracking Example

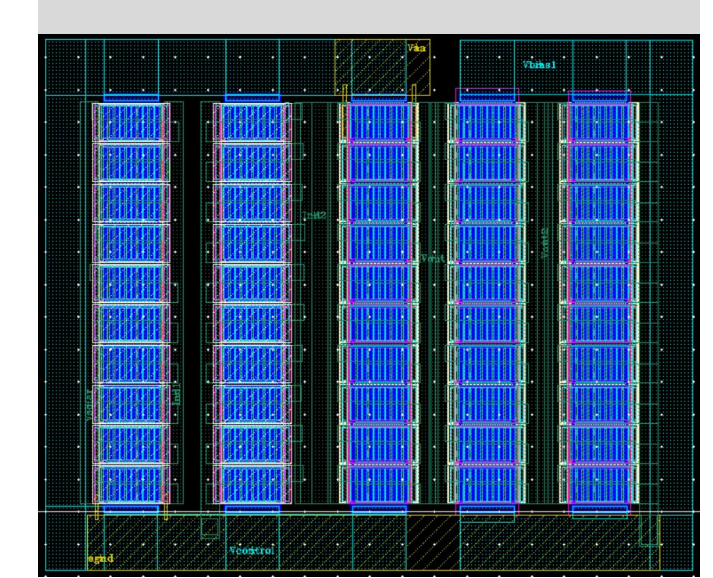
Micro-Solar Manager



True Silicon Size: 2mm x 2mm



Boost Converter



Future Work

Testing This Chip

The Configurability Layer makes CRIMSON exceedingly **easy to test**. 116 scan chain bits and 40 pins of our 64 available can be used to test our chip. Every control signal can be overridden, and all sensor data can be read out through the scan chain. This functionality was given special consideration not only for verification, but also for **hands-on practice** in future testing courses at WPI.

Future Uses in CHIPS Design Studio Projects

CRIMSON was originally pitched as a power controller for a previous CHIPS Design Studio project, the 2025 MQP "Development of a Secure CMOS Image Sensor" by Hubert Liu, Francesco Marrocco, Kelu Liu, Yuran Xue, and Philip Miu. However, we found we could significantly expand the functionality of the chip, allowing it to be used for any future CHIPS Design Studio project that needs reliable power from 3.3V to 1.2V. Additionally, as mentioned in the Purpose, CRIMSON may be deployed in the field where it needs to harvest its own power. Much of the control blocks can be modified through the scan chain, allowing for widely varying needs to be met by the same chip.

References

- [1] W. Kester, "MT-014 TUTORIAL Basic DAC Architectures I: String DACs and Thermometer (Fully Decoded) DACs."
- [2] W. Kester, "MT-016 TUTORIAL Basic DAC Architectures III: Segmented DACs."
- [3] N. Yadav, K. E. Falidas, D. Lehniger, K. Seidel, and T. Kämpfe, "Low-Power and Rail-to-Rail StrongARM DL-Latch Comparator: Design, Analysis and Performance Evaluation," in *International Conference on Modern Circuits and Systems Technologies, MOCAST*, Institute of Electrical and Electronics Engineers Inc., 2025.
- [4] S. U. Ay, "A sub-1 Volt 10-bit supply boosted SAR ADC design in standard CMOS," *Analog Integr. Circuits Signal Process.*, vol. 66, no. 2, pp. 213–221, Feb. 2011.
- [5] "How to Design Flexible Processor Power Systems Using PMICs Wi-Fi is a registered trademark of Wi-Fi Alliance. (1) (2) (3) (4) Application Report How to Design Flexible Processor Power Systems Using PMICs," 2015. [Online]. Available: www.ti.com
- [6] A. Applications Journal, "A DAC for all precision occasions." [Online]. Available: www.ti.com/aaj
- [7] A. Kanago, V. Barry, B. Sprague, I. Cevik and S. Ay, "A low power maximum power point tracker and power management system in 0.5µm CMOS," 2012 IEEE 55th MWSCAS, Boise, ID, USA, 2012, pp. 238–241.